What are the most general values of θ which satisfy the equations:

निम्न समीकरणों के मुख्य सामान्य हल क्या होंगे जो निम्न को सन्तुष्ट करते है :

1. $\sin\theta = \frac{1}{2}$

Ans. $n\pi + (-1)^n \frac{\pi}{6}$

Sol. $\sin\theta = \frac{1}{2}$

$\Rightarrow \sin\theta = \sin\frac{\pi}{6}$

$\Rightarrow \theta = n\pi + (-1)^n \frac{\pi}{6}; n \in I$

2. $\sin\theta = -\frac{\sqrt{3}}{2}$

Ans. $n\pi - (-1)^n \frac{\pi}{3}$

Sol. $\sin\theta = -\frac{\sqrt{3}}{2}$

$\sin\theta = \sin\left(-\frac{\pi}{3}\right)$

$\Rightarrow \theta = n\pi + (-1)^{n+1} \frac{\pi}{3}; n \in I$

3. $\sin\theta = \frac{1}{\sqrt{2}}$

Ans. $n\pi + (-1)^n \frac{\pi}{4}$

Sol. $\sin\theta = \frac{1}{\sqrt{2}}$

$\Rightarrow \sin\theta = \sin\frac{\pi}{4}$

$\Rightarrow \theta = n\pi + (-1)^n \frac{\pi}{4}; n \in I$

4. $\cos\theta = -\frac{1}{2}$

Ans. $2n\pi \pm \frac{2\pi}{3}$

Sol. $\cos\theta = -\frac{1}{2}$

$\Rightarrow \cos\theta = \cos\frac{2\pi}{3}$

$\Rightarrow \theta = 2n\pi \pm \frac{2\pi}{3}; n \in I$

5. $\cos\theta = \frac{\sqrt{3}}{2}$

Ans. $2n\pi \pm \frac{\pi}{6}$

Sol. $\cos\theta = \frac{\sqrt{3}}{2}$

$\Rightarrow \cos\theta = \cos\left(\frac{\pi}{6}\right)$

$\Rightarrow \theta = 2n\pi \pm \frac{\pi}{6}; n \in I$

6. $\cos\theta = -\frac{1}{\sqrt{2}}$

Ans. $2n\pi \pm \frac{3\pi}{4}$

Sol. $\cos\theta = -\frac{1}{\sqrt{2}}$

$\Rightarrow \cos\theta = \cos\left(\frac{3\pi}{4}\right)$

$\Rightarrow \theta = 2n\pi \pm \frac{3\pi}{4}; n \in I$

7. $\tan\theta = \sqrt{3}$

Ans. $n\pi + \frac{\pi}{3}$

Sol. $\tan\theta = \sqrt{3}$

$\Rightarrow \tan\theta = \tan\left(\frac{\pi}{3}\right)$

$\Rightarrow \theta = n\pi + \frac{\pi}{3}; n \in Z$

8. $\tan\theta = -1$

Ans. $n\pi + \frac{3\pi}{4}$

Sol. $\tan\theta = -1$

$\Rightarrow \tan\theta = \tan\left(-\frac{\pi}{4}\right) = \tan\left(\frac{3\pi}{4}\right)$

$\Rightarrow \theta = n\pi + \frac{3\pi}{4}; n \in Z$

9. $\cot\theta = 1$

Ans. $n\pi + \frac{\pi}{4}$

Sol. $\cot\theta = 1$

$\Rightarrow \tan\theta = 1$

$\Rightarrow \theta = n\pi + \frac{\pi}{4}; n \in Z$

10. $\sec\theta = 2$

Ans. $2n\pi \pm \frac{\pi}{3}$

Sol. $\sec\theta = 2$

$\Rightarrow \cos\theta = \frac{1}{2} = \cos\frac{\pi}{3}$

$\Rightarrow \theta = 2n\pi \pm \frac{\pi}{3}; n \in I$

11. $\operatorname{cosec}\theta = \frac{2}{\sqrt{3}}$

Ans. $n\pi + (-1)^n \frac{\pi}{3}$

Sol. $\operatorname{cosec}\theta = \frac{2}{\sqrt{3}}$

$\Rightarrow \sin\theta = \frac{\sqrt{3}}{2} = \sin\left(\frac{\pi}{3}\right)$

$\Rightarrow \theta = n\pi + (-1)^n \frac{\pi}{3}; n \in I$

12. $\sin^2\theta = 1$

Ans. $n\pi \pm \frac{\pi}{2}$

Sol. $\sin^2\theta = 1$

$\Rightarrow \sin^2\theta = \sin^2\frac{\pi}{2}$

$\Rightarrow \theta = n\pi \pm \frac{\pi}{2}; n \in I$

13. $\cos^2\theta = \frac{1}{4}$

Ans. $n\pi \pm \frac{\pi}{3}$

Sol. $\cos^2\theta = \frac{1}{4}$

$\Rightarrow \cos^2\theta = \cos^2\frac{\pi}{3}$

$\Rightarrow \theta = n\pi \pm \frac{\pi}{3}; n \in I$

14. $\tan^2\theta = \frac{1}{3}$

Ans. $n\pi \pm \frac{\pi}{6}$

Sol. $\tan^2\theta = \frac{1}{3}$

$\Rightarrow \tan^2\theta = \tan^2\frac{\pi}{6}$

$\Rightarrow \theta = n\pi \pm \frac{\pi}{6}; n \in I$

15. $4\sin^2\theta = 3$

Ans. $n\pi \pm \frac{\pi}{3}$

Sol. $4\sin^2\theta = 3$

$\Rightarrow \sin^2\theta = \frac{3}{4}$

$\Rightarrow \sin^2\theta = \sin^2\frac{\pi}{3}$

$\Rightarrow \theta = n\pi \pm \frac{\pi}{3}; n \in I$

16. $2\cot^2\theta = \text{cosec}^2\theta$

Ans. $n\pi \pm \frac{\pi}{4}$

Sol. $2\cot^2\theta = \text{cosec}^2\theta$

$\Rightarrow \cos^2\theta = \frac{1}{2}$

$\Rightarrow \cos^2\theta = \cos^2\frac{\pi}{4}$

$\Rightarrow \theta = n\pi \pm \frac{\pi}{4}; n \in I$

17. $\sec^2\theta = \frac{4}{3}$

Ans. $n\pi \pm \frac{\pi}{6}$

Sol. $\sec^2\theta = \frac{4}{3}$

$\Rightarrow \cos^2\theta = \frac{3}{4}$

$\Rightarrow \cos^2\theta = \cos^2\frac{\pi}{6}$

$\Rightarrow \theta = n\pi \pm \frac{\pi}{6}; n \in I$

18. What is the most general value of θ that satisfies both of the equations $\cos\theta = -\frac{1}{\sqrt{2}}$ & $\tan\theta = 1$?

θ का मुख्य सामान्य मान क्या होगा जो समीकरणों $\cos\theta = -\frac{1}{\sqrt{2}}$ तथा $\tan\theta = 1$ को सन्तुष्ट करता है

Ans. $(2n+1)\pi + \frac{\pi}{4}$

Sol. $\cos\theta = -\frac{1}{\sqrt{2}}$

and $\tan\theta = 1$

$\Rightarrow \cos\theta = \cos\frac{3\pi}{4}$

& $\tan\theta = \tan\frac{\pi}{4}$

$\Rightarrow \theta = 2n\pi \pm \frac{3\pi}{4}; n \in I$

& $\theta = n\pi + \frac{\pi}{4}; n \in Z$

$\Rightarrow \theta = \frac{3\pi}{4}, -\frac{3\pi}{4}, \frac{11\pi}{4}, \frac{5\pi}{4}, -\frac{5\pi}{4}, -\frac{11\pi}{4}, \frac{27\pi}{4}, \frac{21\pi}{4}$

& $\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, -\frac{3\pi}{4}, \frac{13\pi}{4}, \frac{17\pi}{4}, \frac{21\pi}{4}$

$\Rightarrow \theta = \frac{5\pi}{4}, -\frac{3\pi}{4}, \frac{21\pi}{4}, ...$

$\Rightarrow \theta = (2n+1)\pi + \frac{\pi}{4}; n \in I$

19. What is the most general value of θ that satisfies both of the equations $\cot\theta = -\sqrt{3}$ & $\text{cosec}\,\theta = -2$?

θ का मुख्य सामान्य मान क्या होगा जो समीकरणों $\cot\theta = -\sqrt{3}$ तथा $\text{cosec}\,\theta = -2$ को सन्तुष्ट करता है

Ans. $2n\pi - \frac{\pi}{6}$

Sol. $\cot\theta = -\sqrt{3}$ & $\text{cosec}\,\theta = -2$

$\Rightarrow \tan\theta = -\frac{1}{\sqrt{3}}$ & $\sin\theta = -\frac{1}{2}$

In $[0, 2\pi]$

$\theta = \frac{5\pi}{6}, \frac{11\pi}{6}$ & $\theta = \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$

$\Rightarrow \theta = 2n\pi + \frac{11\pi}{6}$

$\Rightarrow \theta = (2n+2)\pi - \frac{\pi}{6}; n \in I$

$\Rightarrow \theta = 2m\pi - \frac{\pi}{6}; m \in I$

20. Find the angles between 0° and 360° which have respectively

(1) Their sines equal to $\frac{\sqrt{3}}{2}$ (2) Their cosines equal to $-\frac{1}{2}$ (3) Their tangents equal to $\frac{1}{\sqrt{3}}$

0° तथा 360° के मध्य कोणों (θ) को ज्ञात करो जबकि

(1) $\sin\theta = \frac{\sqrt{3}}{2}$ (2) $\cos\theta = -\frac{1}{2}$ (3) $\tan\theta = \frac{1}{\sqrt{3}}$

Ans. (1) 60° and 120° (2) 120° and 240° (3) 30° and 210°

Sol. $\theta \in (0, 2\pi)$

(1) $\sin\theta = \frac{\sqrt{3}}{2}$

$\theta = \frac{\pi}{3}, \frac{2\pi}{3}$

(2) $\cos\theta = -\frac{1}{\sqrt{2}}$

$\theta = \frac{4\pi}{3}, \frac{2\pi}{3}$

(3) $\tan\theta = \frac{1}{\sqrt{3}}$

$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$

21. Taking into consideration only angles between 0º and 180º, how many values of x are there if
(1) $\sin x = \frac{5}{7}$, (2) $\cos x = \frac{1}{5}$, (3) $\cos x = -\frac{4}{5}$, (4) $\tan x = \frac{2}{3}$ and (5) $\cot x = -7$
0º तथा 180º के मध्य x के कितने मान होंगे जबकि
(1) $\sin x = \frac{5}{7}$, (2) $\cos x = \frac{1}{5}$, (3) $\cos x = -\frac{4}{5}$, (4) $\tan x = \frac{2}{3}$ तथा (5) $\cot x = -7$

Ans. (1) 2 (2) 1 (3) 1 (4) 1 (5) 1

Sol. $\theta \in (0, \pi)$

(1) $\sin x = \frac{5}{7}$

$\Rightarrow$ There are 2 values of x s.t.

$\sin x = \frac{5}{7}$

(2) $\cos x = \frac{1}{5}$

$\Rightarrow$ There are one value of x s.t. $\cos x = \frac{1}{5}$

(3) $\cos x = -\frac{4}{5}$

$\Rightarrow$ There are 1 value of x in $(0, \pi)$ s.t.

$\cos x = -\frac{4}{5}$

(4) $\tan x = \frac{2}{3}$

$\Rightarrow$ There are 1 value of x in $(0, \pi)$ s.t.

$\tan x = \frac{2}{3}$

(5) $\cot x = -7$

$\Rightarrow \tan x = \frac{-1}{7}$

$\Rightarrow$ There are 1 value of x in $(0, \pi)$ s.t.
$\cot x = -7$

22. Show that the same angles are indicated by the two following formulae

(1) $(2n-1)\frac{\pi}{2} + (-1)^n \frac{\pi}{3}$, (2) $2n\pi \pm \frac{\pi}{6}$, n being any integer.

सिद्ध कीजिये कि नीचे दिये गये सूत्र दो समान कोणों को निरूपित करते है।

(1) $(2n-1)\frac{\pi}{2} + (-1)^n \frac{\pi}{3}$, (2) $2n\pi \pm \frac{\pi}{6}$, n being any integer.

Ans.

Sol. Let $\theta = (2n-1)\frac{\pi}{2} + (-1)^n \frac{\pi}{3}; n \in I$ & $\phi = 2n\pi \pm \frac{\pi}{6}; n \in I$

$\Rightarrow \theta = n\pi - \frac{\pi}{2} + (-1)^n \frac{\pi}{3}$ $\Rightarrow \phi = \frac{\pi}{6}, -\frac{\pi}{6}$

$\Rightarrow \cos\phi = \frac{\sqrt{3}}{2}$

$\Rightarrow \theta + \frac{\pi}{2} = n\pi + (-1)^n \frac{\pi}{3}$

$\Rightarrow \sin\left(\theta + \frac{\pi}{2}\right) = \sin\left(\frac{\pi}{3}\right)$

$\Rightarrow \cos\theta = \frac{\sqrt{3}}{2} = \cos\phi$

23. Prove that the two formulae

(1) $\left(2n + \frac{1}{2}\right)\pi \pm \alpha$ and (2) $n\pi + (-1)^n \left(\frac{\pi}{2} - \alpha\right)$ denote the same angles, n being any integer.

सिद्ध कीजिये

(1) $\left(2n + \frac{1}{2}\right)\pi \pm \alpha$ तथा (2) $n\pi + (-1)^n \left(\frac{\pi}{2} - \alpha\right)$ समान कोणों को निरूपित करते है। जहाँ n पूर्णांक है।

Ans.

Sol. $\left(2n + \frac{1}{2}\right)\pi \pm \alpha$ & $n\pi + (-1)^n \left(\frac{\pi}{2} - \alpha\right)$

Proof :

Let $\theta = 2n\pi + \frac{\pi}{2} \pm \alpha$ & $\phi = n\pi + (-1)^n \left(\frac{\pi}{2} - \alpha\right)$

$\Rightarrow \cos\left(\theta-\frac{\pi}{2}\right)=\cos\alpha$ & $\sin\phi=\sin\left(\frac{\pi}{2}-\alpha\right)$

$\Rightarrow \sin\theta=\cos\alpha$ & $\sin\phi=\cos\alpha$

$\Rightarrow \sin\theta=\sin\phi=\cos\alpha$

24. If $\theta-\alpha=n\pi+(-1)^n\beta$, prove that $\theta=2m\pi+\alpha+\beta$ or else that $\theta=(2m+1)\pi+\alpha-\beta$, where m and n are any integers.

यदि $\theta-\alpha=n\pi+(-1)^n\beta$, सिद्ध कीजिये कि $\theta=2m\pi+\alpha+\beta$ या $\theta=(2m+1)\pi+\alpha-\beta$, जहां m तथा n पूर्णांक है।

Ans.

Sol. Given $\theta-\alpha=n\pi+(-1)^n\beta$

$\Rightarrow \sin(\theta-\alpha)=\sin\beta$

$\Rightarrow \sin\theta\cos\alpha-\cos\theta\sin\alpha=\sin\beta$

Case–I :- If n is even then let $n=2m$

$\Rightarrow \theta=\alpha+2m\pi+\beta$

$\Rightarrow \theta-\alpha-\beta=2m\pi;\ m\in I$

Case–II :- If n is odd then let $n=2m+1$

$\Rightarrow \theta=(2m+1)\pi+\alpha-\beta;\ m\in I$

25. If $\cos p\theta+\cos q\theta=0$, prove that the different values of θ from two arithmetical progressions in which the common differences are $\frac{2\pi}{p+q}$ and $\frac{2\pi}{p-q}$ respectively.

यदि $\cos p\theta+\cos q\theta=0$, तो सिद्ध कीजिये कि θ के भिन्न–भिन्न मान दो स.श्रे. बनाते है जिनका सार्वअन्तर क्रमशः $\frac{2\pi}{p+q}$ तथा $\frac{2\pi}{p-q}$ है।

Ans.

Sol. $\cos p\theta+\cos q\theta=0$

$\Rightarrow 2\cos\left(\frac{p\theta+q\theta}{2}\right)\cos\left(\frac{p\theta-q\theta}{2}\right)=0$

$\Rightarrow \cos\left(\frac{(p+q)\theta}{2}\right)=0$ or $\cos\left(\frac{(p-q)\theta}{2}\right)=0$

$\Rightarrow \frac{(p+q)}{2}\theta=2n\pi\pm\frac{\pi}{2}$ or $\frac{(p-q)\theta}{2}=2m\pi\pm\frac{\pi}{2}$

$\Rightarrow (p+q)\theta=4n\pi\pm\pi$ or $(p-q)\theta=4m\pi\pm\pi$

$\Rightarrow \frac{(p+q)\theta}{2}=(2n+1)\frac{\pi}{2}$ or $\frac{(p-q)\theta}{2}=(2m+1)\frac{\pi}{2}$

$\Rightarrow \theta = \dfrac{(2n+1)\pi}{P+Q}; m \in I$ or $\theta = \dfrac{(2m+1)\pi}{P-Q}; m \in I$

$\theta = \left\{\dfrac{2\pi}{p+q}, \dfrac{3\pi}{P+Q}, \dfrac{5\pi}{P+Q}, ...\right\}$ or $\theta = \left\{....., \dfrac{\pi}{P-Q}, \dfrac{3\pi}{P-Q}, \dfrac{5\pi}{P-Q}, \right\}$

This an A.P. with $d = \dfrac{2\pi}{P+Q}$ This is an A.P. with $d = \dfrac{2\pi}{P-Q}$

Solve the equations

26. $\sin 5\theta = \dfrac{1}{\sqrt{2}}$

Ans. $\dfrac{n\pi}{5} + (-1)^n \dfrac{\pi}{20}$

Sol. $\sin 5\theta = \dfrac{1}{\sqrt{2}}$

$\Rightarrow \sin 5\theta = \sin\left(\dfrac{\pi}{4}\right)$

$\Rightarrow 5\theta = n\pi + (-1)^n \dfrac{\pi}{4}; n \in I$

$\Rightarrow \theta = \dfrac{n\pi}{5} + (-1)^n \dfrac{\pi}{20}; n \in I$

27. $3\tan^2\theta = 1$

Ans. $n\pi \pm \dfrac{\pi}{6}$

Sol. $3\tan^2\theta = 1$

$\Rightarrow \tan^2\theta = \dfrac{1}{3}$

$\Rightarrow \tan^2\theta = \tan^2\dfrac{\pi}{6}$

$\Rightarrow \theta = n\pi \pm \dfrac{\pi}{6}; n \in I$

28. $\sin(\theta - \phi) = \dfrac{1}{2}$ and $\cos(\theta + \phi) = \dfrac{1}{2}$

$\sin(\theta - \phi) = \dfrac{1}{2}$ तथा $\cos(\theta + \phi) = \dfrac{1}{2}$

Ans. $\theta = \left(m + \frac{n}{2}\right)\pi \pm \frac{\pi}{6} + (-1)^n \frac{\pi}{12}; \phi = \left(m - \frac{n}{2}\right)\pi \pm \frac{\pi}{6} - (-1)^n \frac{\pi}{12}$

Sol. $\sin(\theta - \phi) = \frac{1}{2}$ and $\cos(\theta + \phi) = \frac{1}{2}$

$\Rightarrow \theta - \phi = n\pi + (-1)^n \frac{\pi}{6}$ and $\theta + \phi = 2m\pi \pm \frac{\pi}{3}$

$\Rightarrow 2\theta = n\pi + 2m\pi \pm \frac{\pi}{3} + (-1)^n \frac{\pi}{6}$

$\Rightarrow \theta = \left(m + \frac{n}{2}\right)\pi \pm \frac{\pi}{6} + (-1)^n \frac{\pi}{12}$

& $2\phi = (2m - n)\pi \pm \frac{\pi}{3} + (-1)^{n+1} \frac{\pi}{6}$

$\Rightarrow \phi = \left(m - \frac{n}{2}\right)\pi \pm \frac{\pi}{6} - (-1)^n \frac{\pi}{12}$

29. $\cos(2x + 3y) = \frac{1}{2}$ and $\cos(3x + 2y) = \frac{\sqrt{3}}{2}$

$\cos(2x + 3y) = \frac{1}{2}$ तथा $\cos(3x + 2y) = \frac{\sqrt{3}}{2}$

Ans. $\frac{1}{5}\left[(6m - 4n)\pi \pm \frac{\pi}{2} \mp \frac{2\pi}{3}\right]; \frac{1}{5}\left[(6n - 4m)\pi \pm \pi \mp \frac{\pi}{3}\right]$

Sol. $\cos(2x + 3y) = \frac{1}{2}$ and $\cos(3x + 2y) = \frac{\sqrt{3}}{2}$

$\Rightarrow 2x + 3y = 2n\pi \pm \frac{\pi}{3}$ & $3x + 2y = 2m\pi \pm \frac{\pi}{6}$

$x + y = (n + m)\frac{2\pi}{5} \pm \frac{\pi}{15} \pm \frac{\pi}{30}$

$x - y = (m - n)2\pi \pm \frac{\pi}{6} \mp \frac{\pi}{3}$

$\Rightarrow 2x = \left(\frac{n}{5} + \frac{m}{5} + m - n\right)2\pi \pm \left(\frac{\pi}{15} - \frac{\pi}{3}\right) \pm \left(\frac{\pi}{6} + \frac{\pi}{30}\right)$

$\Rightarrow x = \frac{1}{5}\left[(6m - 4n)\pi \pm \frac{\pi}{2} \mp \frac{2\pi}{3}\right]$

& $y = \frac{1}{5}\left[(6n - 4m)\pi \pm \pi \mp \frac{\pi}{3}\right]$

1. $\cos 2\theta = \cos^2 \theta$

Ans. $n\pi$

Sol. Given, $\cos 2\theta = \cos^2 \theta$

$2\cos^2 \theta - 1 = \cos^2 \theta$

$\Rightarrow \quad \cos^2 \theta - 1 = 0$

$\Rightarrow \quad \cos^2 \theta = 1$

$\Rightarrow \quad \cos\theta = \pm 1$

$\Rightarrow \quad \theta = n\pi; n \in I$

2. $4 \cos \theta - 3 \sec \theta = \tan \theta$

Ans. $\sin\theta = \dfrac{\pm\sqrt{17}-1}{8}$

Sol. Given, $4 \cos \theta - 3 \sec \theta = \tan \theta$

$\Rightarrow \quad 4\cos\theta - \dfrac{3}{\cos\theta} = \dfrac{\sin\theta}{\cos\theta}$

$\Rightarrow \quad 4 \cos^2 \theta - 3 = \sin \theta \quad \{\because \cos\theta \neq 0\}$

$\Rightarrow \quad 4 - 4 \sin^2 \theta - 3 = \sin \theta$

$\Rightarrow \quad 4\sin^2 \theta + \sin \theta - 1 = 0$

$\therefore \quad \sin \theta = \dfrac{-1 \pm \sqrt{(1+16)}}{8} = \dfrac{\pm\sqrt{17}-1}{8}$

3. $\cos 2\theta + 3 \cos \theta = 0$

Ans. $\cos\theta = \dfrac{\sqrt{17}-3}{4}$

Sol. Given, $\cos 2\theta + 3 \cos \theta = 0$

$\Rightarrow \quad 2\cos^2\theta + 3 \cos \theta - 1 = 0$

$\Rightarrow \quad \cos\theta = \dfrac{-3 \pm \sqrt{(9+8)}}{4}$

$\therefore \quad \cos\theta = \dfrac{\sqrt{17}-3}{4} \text{ or } \dfrac{-\sqrt{17}-3}{4}$

But is impossible, as $\cos\theta \leq -1$

$\therefore \cos\theta = \dfrac{\sqrt{17}-3}{4}$

4. $\cos 3\theta + 2\cos\theta = 0$

Ans. $n\pi \pm \dfrac{\pi}{3}$ or $n\pi + \dfrac{\pi}{2}$

Sol. Given, $\cos 3\theta + 2\cos\theta = 0$

$\Rightarrow 4\cos^3\theta - 3\cos\theta + 2\cos\theta = 0$

$\Rightarrow 4\cos^3\theta - \cos\theta = 0$

$\Rightarrow \cos\theta\,(4\cos^2\theta - 1) = 0$

$\Rightarrow \cos\theta = 0$ or $\cos^2\theta = \dfrac{1}{4}$

$\Rightarrow \cos\theta = 0$ or $\cos^2\theta = \left(\dfrac{1}{2}\right)^2$

$\Rightarrow \cos\theta = 0$ or $\cos^2\theta = \cos^2\dfrac{\pi}{3}$

$\Rightarrow \theta = n\pi + \dfrac{\pi}{2}; n \in I$ or $\theta = n\pi \pm \dfrac{\pi}{3}; n \in I$

5. $\cos 2\theta = (\sqrt{2}+1)\left(\cos\theta - \dfrac{1}{\sqrt{2}}\right)$

Ans. $2n\pi \pm \dfrac{\pi}{3}; 2n\pi \pm \dfrac{\pi}{4}$

Sol. Given, $\cos 2\theta = (\sqrt{2}+1)\left(\cos\theta - \dfrac{1}{\sqrt{2}}\right)$

$\Rightarrow 2\cos^2\theta - 1 = \left[\sqrt{2}+1\right]\left(\cos\theta - \dfrac{1}{\sqrt{2}}\right)$

$2\cos^2\theta - \left[\sqrt{2}+1\right]\cos\theta - 1 + 1 + \dfrac{1}{\sqrt{2}} = 0$

$\Rightarrow \sqrt{2}\cos\theta\left[\sqrt{2}\cos\theta - 1\right] - \dfrac{1}{\sqrt{2}}\left[\sqrt{2}\cos\theta - 1\right] = 0$

$\Rightarrow \left[\left(\sqrt{2}\cos\theta - 1\right)\right]\left(\sqrt{2}\cos\theta - \dfrac{1}{\sqrt{2}}\right) = 0$

$\Rightarrow \sqrt{2}\cos\theta - \dfrac{1}{\sqrt{2}} = 0$ or

or $\sqrt{2}\cos\theta - 1 = 0$

If $\sqrt{2}\cos\theta - \frac{1}{\sqrt{2}} = 0$

Then $\cos\theta = \frac{1}{2}$ or $\theta = \frac{\pi}{3}$

$\therefore$ G.V. of $\theta = 2n\pi \pm \frac{\pi}{3}; n \in I$

If $\sqrt{2}\cos\theta - 1 = 0$

Then $\cos\theta = \frac{1}{\sqrt{2}}$ or $\theta = \frac{\pi}{4}$

$\therefore$ G.V. of $\theta = 2n\pi \pm \frac{\pi}{4}; n \in I$

6. $\cot\theta - \tan\theta = 2$

Ans. $\left(n + \frac{1}{4}\right)\frac{\pi}{2}; n \in I$

Sol. Given, $\cot\theta - \tan\theta = 2$

$\Rightarrow \frac{\cos\theta}{\sin\theta} - \frac{\sin\theta}{\cos\theta} = 2$

$\Rightarrow \frac{\cos^2\theta - \sin^2\theta}{\sin\theta\cos\theta} = 2$

$\Rightarrow \cos 2\theta = \sin 2\theta$

$\Rightarrow \tan 2\theta = 1$

$\Rightarrow \tan 2\theta = 1$ or $\tan 2\theta = \tan\frac{\pi}{4}$

G.V. of $2\theta = n\pi + \frac{\pi}{4}$

$\Rightarrow$ G.V. of $\theta = \frac{\pi}{2}\left(n + \frac{1}{4}\right); n \in I$

7. $\cos^2\theta - \sin\theta - \frac{1}{4} = 0$

Ans. $n\pi + (-1)^n \frac{\pi}{6}; n \in I$

Sol. $\cos^2\theta - \sin\theta - \frac{1}{4} = 0$

$\Rightarrow 1 - \sin^2\theta - \sin\theta - \frac{1}{4} = 0$

$\Rightarrow \sin^2\theta + \sin\theta = \frac{3}{4}$

$\Rightarrow 4\sin^2\theta + 4\sin\theta - 3 = 0$

$\Rightarrow \left(\sin\theta - \frac{1}{2}\right)\left(\sin\theta + \frac{3}{2}\right)\cdot 4 = 0$

$\Rightarrow \sin\theta = \frac{1}{2}$ or $\underbrace{\sin\theta = \frac{-3}{2}}_{\text{Not possible}}$

$\Rightarrow \theta = n\pi + (-1)^n \cdot \frac{\pi}{6}; n \in I$

8. $2\cos^2\theta + 3\cos\theta = 0$

Ans. $(2n+1)\frac{\pi}{2}; n \in I$

Sol. $2\cos^2\theta + 3\cos\theta = 0$

$\cos\theta(2\cos\theta + 3) = 0$

$\because\ 2\cos\theta + 3 \neq 0$

$\Rightarrow \cos\theta = 0$

$\Rightarrow \theta = (2n+1)\frac{\pi}{2}; n \in I$

9. $2\sqrt{3}\cos^2\theta = \sin\theta$

Ans. $n\pi + (-1)^n \frac{\pi}{3}; n \in I$

Sol. $2\sqrt{3}\cos^2\theta = \sin\theta$

$2\sqrt{3}(1 - \sin^2\theta) = \sin\theta$

$\Rightarrow 2\sqrt{3}\sin^2\theta + \sin\theta - 2\sqrt{3} = 0$

$\sin\theta = \frac{-1 \pm \sqrt{1+48}}{4\sqrt{3}}$

$\Rightarrow \underbrace{\sin\theta = -\frac{2}{\sqrt{3}}}_{\text{Not possible}}$ or $\sin\theta = \frac{\sqrt{3}}{2}$

$\Rightarrow \sin\theta = \frac{\sqrt{3}}{2}$

$\Rightarrow \theta = n\pi + (-1)^n \frac{\pi}{3}$

10. $\cos\theta + \cos^2\theta = 1$

Ans. $\cos\theta = \frac{\sqrt{5}-1}{2}$

Sol. $\cos\theta + \cos^2\theta = 1$

$\Rightarrow \cos^2\theta + \cos\theta - 1 = 0$

$\cos\theta = \frac{-1 \pm \sqrt{5}}{2}$

$\Rightarrow \cos\theta = \frac{-1-\sqrt{5}}{2}$ } Not possible

or $\cos\theta = \frac{\sqrt{5}-1}{2}$

$\Rightarrow \cos\theta = \frac{\sqrt{5}-1}{2}$

11. $4\cos\theta - 3\sec\theta = 2\tan\theta$

Ans. $n\pi + (-1)^n \frac{\pi}{10}; n \in I$ or $n\pi - (-1)^n \frac{3\pi}{10}; n \in I$

Sol. $4\cos\theta - 3\sec\theta = 2\tan\theta$

$4\cos^2\theta - 3 = 2\sin\theta$ ($\because \cos\theta \neq 0$)

$\Rightarrow 4(1 - \sin^2\theta) - 3 = 2\sin\theta$

$\Rightarrow 4\sin^2\theta + 2\sin\theta - 1 = 0$

$\Rightarrow \sin\theta = \frac{-2 \pm \sqrt{20}}{8}$

$\Rightarrow \sin\theta = \frac{\sqrt{5}-1}{4}$ or $\sin\theta = \frac{-1-\sqrt{5}}{4}$

$\Rightarrow \theta = n\pi + (-1)^n \frac{\pi}{10}$ or $\theta = n\pi + (-1)^{n+1} \frac{3\pi}{10}$

12. $\sin^2\theta - 2\cos\theta + \frac{1}{4} = 0$

Ans. $\theta = 2n\pi \pm \frac{\pi}{3}$

Sol. $\sin^2\theta - 2\cos\theta + \frac{1}{4} = 0$

$\Rightarrow 1 - \cos^2\theta - 2\cos\theta + \frac{1}{4} = 0$

$\Rightarrow 4\cos^2\theta + 8\cos\theta - 5 = 0$

$\Rightarrow \cos\theta = \frac{-8 \pm \sqrt{64+80}}{8}$

$\Rightarrow \cos\theta = \frac{1}{2}$ or $\underbrace{\cos\theta = -\frac{5}{2}}_{\text{Not possible}}$

$\Rightarrow \cos\theta = \frac{1}{2}$

$\Rightarrow \theta = 2n\pi \pm \frac{\pi}{3}; n \in I$

13. $\tan^2\theta - (1 + \sqrt{3})\tan\theta + \sqrt{3} = 0$

Ans. $\theta = n\pi + \frac{\pi}{4}$ or $n\pi + \frac{\pi}{3}; n \in I$

Sol. $\tan^2\theta - \left(1+\sqrt{3}\right)\tan\theta + \sqrt{3} = 0$

$\Rightarrow \tan\theta = \frac{1+\sqrt{3} \pm \sqrt{4+2\sqrt{3}-4\sqrt{3}}}{2}$

$\Rightarrow \tan\theta = 1$ or $\tan\theta = \sqrt{3}$

$\Rightarrow \theta = n\pi + \frac{\pi}{4}; n \in I$ or $\theta = n\pi + \frac{\pi}{3}; n \in I$

14. $\cot^2\theta + \left(\sqrt{3} + \frac{1}{\sqrt{3}}\right)\cot\theta + 1 = 0$

Ans. $\theta = n\pi + \frac{2\pi}{3}$ or $n\pi + \frac{5\pi}{6}$

Sol. $\cot^2\theta+\left(\sqrt{3}+\frac{1}{\sqrt{3}}\right)\cot\theta+1=0$

$\Rightarrow \cot\theta\left(\cot\theta+\sqrt{3}\right)+\frac{1}{\sqrt{3}}\left(\cot\theta+\sqrt{3}\right)=0$

$\Rightarrow \cot\theta=-\sqrt{3}$ or $\cot\theta=\frac{-1}{\sqrt{3}}$

$\Rightarrow \tan\theta=\frac{-1}{\sqrt{3}}$ or $\tan\theta=-\sqrt{3}$

$\Rightarrow \theta=n\pi+\frac{2\pi}{3}$ or $\theta=n\pi+\frac{5\pi}{6}$

15. $\cot\theta - ab\tan\theta = a - b$

Ans. $\tan\theta=\frac{1}{a}\text{ or }-\frac{1}{b}$

Sol. $\cot\theta-ab\tan\theta=a-b$

$\Rightarrow 1-ab\tan^2\theta=\left(a-b\right)\tan\theta$

$\Rightarrow ab\tan^2\theta+\left(a-b\right)\tan\theta-1=0$

$\Rightarrow \tan\theta=\frac{b-a\pm\sqrt{a^2+b^2-2ab+4ab}}{2ab}$

$\Rightarrow \tan\theta=\frac{b-a\pm\left(a+b\right)}{2ab}$

$\Rightarrow \tan\theta=\frac{1}{a}$ or $\tan\theta=\frac{-1}{b}$

16. $\tan^2\theta + \cot^2\theta = 2$

Ans. $\theta=n\pi\pm\frac{\pi}{4}$

Sol. $\tan^2\theta + \cot^2\theta = 2$

$\Rightarrow \sin^4\theta + \cos^4\theta = 2\sin^2\theta\cos^2\theta$

$\Rightarrow (\sin^2\theta - \cos^2\theta)^2 = 0$

$\Rightarrow \tan^2\theta = 1$

$\Rightarrow \theta=n\pi\pm\frac{\pi}{4};n\in I$

17. $3(\sec^2\theta + \tan^2\theta) = 5$

Ans. $n\pi \pm \frac{\pi}{6}$

Sol. $3\,(\sec^2\theta + \tan^2\theta) = 5$

$\Rightarrow 3\,(1 + \sin^2\theta) = 5\cos^2\theta$

$\Rightarrow 3\,(1 + 1 - \cos^2\theta) = 5\cos^2\theta$

$\Rightarrow 6 = 8\cos^2\theta$

$\Rightarrow \cos^2\theta = \frac{3}{4}$

$\Rightarrow \cos^2\theta = \left(\frac{\sqrt{3}}{2}\right)^2 = \cos^2\frac{\pi}{6}$

$\Rightarrow \theta = n\pi \pm \frac{\pi}{6}; n \in I$

18. $\cot\theta + \tan\theta = 2\,\text{cosec}\,\theta$

Ans. $2n\pi \pm \frac{\pi}{3}$

Sol. $\cot\theta + \tan\theta = 2\,\text{cosec}\,\theta$

$\Rightarrow \frac{\cos^2\theta + \sin^2\theta}{\cos\theta\sin\theta} = \frac{2}{\sin\theta}$

$\because \sin\theta \neq 0$

$\Rightarrow \cos\theta = \frac{1}{2}$

$\Rightarrow \theta = 2n\pi \pm \frac{\pi}{3}; n \in I$

19. $4\cos^2\theta + \sqrt{3} = 2(\sqrt{3} + 1)\cos\theta$

Ans. $2n\pi \pm \frac{\pi}{3}$ or $2n\pi \pm \frac{\pi}{6}$

Sol. $4\cos^2\theta + \sqrt{3} = 2\left(\sqrt{3} + 1\right)\cos\theta$

$\Rightarrow 4\cos^2\theta - 2\left(\sqrt{3} + 1\right)\cos\theta + \sqrt{3} = 0$

$\Rightarrow \cos\theta = \frac{2\left(\sqrt{3} + 1\right) \pm \sqrt{4\left(\sqrt{3} + 1\right)^2 - 16\sqrt{3}}}{8}$

$$= \frac{2\left(\sqrt{3}+1\right) \pm \sqrt{16-8\sqrt{3}}}{8}$$

$$= \frac{2\left(\sqrt{3}+1\right) \pm \sqrt{4\left(\sqrt{3}-1\right)^2}}{8}$$

$$= \frac{2\left(\sqrt{3}+1\right) \pm 2\left(\sqrt{3}-1\right)}{8}$$

$\Rightarrow \cos\theta = \frac{\sqrt{3}}{2}$ or $\cos\theta = \frac{1}{2}$

$\Rightarrow \theta = 2n\pi \pm \frac{\pi}{6}$ or $\theta = 2n\pi \pm \frac{\pi}{3}$

20. $3\sin^2\theta - 2\sin\theta = 1$

Ans. $\sin\theta = 1$ or $-\frac{1}{3}$

Sol. $3\sin^2\theta - 2\sin\theta = 1$

$$\sin\theta = \frac{2 \pm \sqrt{4+12}}{6}$$

$\Rightarrow \sin\theta = 1$ or $\sin\theta = \frac{-1}{3}$

21. $\sqrt{3}\cos\theta + \sin\theta = \sqrt{2}$

Ans. $n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{3}$

Sol. Dividing the equation by 2, we get

$$\frac{\sqrt{3}}{2}\cos\theta + \frac{1}{2}\sin\theta = \frac{1}{\sqrt{2}}$$

$\Rightarrow \sin 60^\circ \cos\theta + \cos 60^\circ \sin\theta = \frac{1}{\sqrt{2}}$

$\Rightarrow \sin\left(\frac{\pi}{3}+\theta\right) = \sin 45^\circ$

$\Rightarrow \left(\frac{\pi}{3}+\theta\right) = n\pi + (-1)^n \frac{\pi}{4}$

$\Rightarrow \theta = n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{3}; n \in I$

22. $\sin\theta + \cos\theta = \sqrt{2}$

Ans. $2n\pi + \frac{\pi}{4}$

Sol. Dividing the equation by $\sqrt{2}$, we get

$\frac{1}{\sqrt{2}}\sin\theta + \frac{1}{\sqrt{2}}\cos\theta = 1$

$\Rightarrow \cos 45^\circ \cos\theta + \sin 45^\circ \sin\theta = 1$

$\Rightarrow \cos(\theta - 45^\circ) = 1$

$\Rightarrow \theta - \frac{\pi}{4} = 2n\pi$

$\Rightarrow \theta = 2n\pi + \frac{\pi}{4}; n \in I$

23. $\sqrt{3}\ \sin\theta - \cos\theta = \sqrt{2}$

Ans. $n\pi + \frac{\pi}{6} + (-1)^n \frac{\pi}{4}$

Sol. Dividing the equatin by 2

$\frac{\sqrt{3}}{2}\sin\theta - \frac{1}{2}\cos\theta = \frac{1}{\sqrt{2}}$

$\Rightarrow \sin\theta\cos\frac{\pi}{6} - \sin\frac{\pi}{6}\cos\theta = \sin\frac{\pi}{4}$

$\Rightarrow \sin\left(\theta - \frac{\pi}{6}\right) = \sin\frac{\pi}{4}$

$\Rightarrow \theta - \frac{\pi}{6} = n\pi + (-1)^n \frac{\pi}{4}$

$\Rightarrow \theta = n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{6}$

24. $\sin x + \cos x = \sqrt{2}\ \cos A$

Ans. $2n\pi + \frac{\pi}{4} \pm A$

Sol. Dividing the equation by $\sqrt{2}$, we get

$$\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{2}}\cos x = \cos A$$

$$\Rightarrow \cos\left(x - \frac{\pi}{4}\right) = \cos A$$

$$\Rightarrow x - \frac{\pi}{4} = 2n\pi \pm A$$

$$\Rightarrow x = 2n\pi. \pm A + \frac{\pi}{4}$$

25. $5\sin\theta + 2\cos\theta = 5$ (given $\tan 21°48' = 0.4$)

Ans. $-21°48' + n\,180° + (-1)^n [68°12']$

Sol. Dividing the equation by $\sqrt{(29)}$, we get

$$\Rightarrow \frac{5}{\sqrt{(29)}}\sin\theta + \frac{2}{\sqrt{(29)}}\cos\theta = \frac{5}{\sqrt{(29)}}$$

Now if $\sin\alpha = \frac{2}{\sqrt{(29)}}, \cos\theta = \frac{5}{\sqrt{(29)}}$

then $\cos\alpha\sin\theta + \sin\alpha\cos\theta = \cos\alpha$

$$\Rightarrow \sin(\alpha + \theta) = \sin(90° - \alpha) = \sin\left(\frac{\pi}{2} - \alpha\right)$$

$$\Rightarrow \alpha + \theta = n\pi + (-1)^n\left(\frac{\pi}{2} - \alpha\right)$$

$$\Rightarrow \theta = n\pi + (-1)^n\left(\frac{\pi}{2} - \alpha\right) - \alpha$$

where $\tan\alpha = \frac{2}{5}$

$$\Rightarrow \alpha = 21°\,48'$$

$\therefore\ \theta = -21^\circ 48' + n\,180^\circ + (-1)^n\,[68^\circ 12']$

26. $6\cos x + 8\sin x = 9$ (given $\tan 53^\circ 8' = 1\frac{1}{3}$ and $\cos 25^\circ 50' = 0.9$)

Ans. $2n\pi + 78^\circ 58'; 2n\pi + 27^\circ 18'$

Sol. Dividing the equation by 10, we get

$$\frac{3}{5}\cos x + \frac{4}{5}\sin x = \frac{9}{10}$$

$\Rightarrow \cos 53^\circ 8' \cos x + \sin 53^\circ 8' \sin x = \cos 25^\circ 50'$

$\Rightarrow \cos(x - 53^\circ 8') = \cos 25^\circ 50'$

$\Rightarrow x - 53^\circ 8'\ 2n\pi \pm 25^\circ 50'$

$\Rightarrow x = 2n\pi \pm 25^\circ 50' + 53^\circ 8';\ n \in I$

$\Rightarrow x = 2n\pi + 27^\circ 18', 2n\pi + 78^\circ 58';\ n \in I$

27. $1 + \sin^2\theta = 3\sin\theta\cos\theta$ (given $\tan 71^\circ 34' = 3$)

Ans. $n.180^\circ + 45^\circ; n.180^\circ + 26^\circ 34'$**;** $n \in I$

Sol. $$1 + \left(\frac{1-\cos 2\theta}{2}\right) = \frac{3}{2}\sin 2\theta$$

$$\Rightarrow \frac{3}{2} - \frac{1}{2}\cos 2\theta = \frac{3}{2}\sin 2\theta$$

$\Rightarrow 3\sin 2\theta + \cos 2\theta = 3$

Dividing by $\sqrt{(10)}$, we get

$$\frac{3}{\sqrt{(10)}}\sin 2\theta + \frac{1}{\sqrt{(10)}}\cos 2\theta = \frac{3}{\sqrt{(10)}}$$

$$\Rightarrow \sin 18^\circ 34' \sin 2\theta + \cos 71^\circ 34' \cos 2\theta = \frac{3}{\sqrt{(10)}}$$

$\Rightarrow \cos(2\theta - 71^\circ 34') = \sin 71^\circ 34' = \cos 18^\circ 26'$

$\Rightarrow 2\theta - 71^\circ 34' = 2n\pi \pm 18^\circ 26';\ n \in I$

$$\Rightarrow \theta = \frac{1}{2}\left(2n\pi \pm 18^\circ 26' + 71^\circ 34'\right);\ n \in I$$

$\Rightarrow \theta = n\pi + 45^\circ, n\pi + 26^\circ 34'; n \in I$

28. $\operatorname{cosec}\theta = \cot\theta + \sqrt{3}$

Ans. $2n\pi + \frac{2\pi}{3}; n \in I$

Sol. Given, $\operatorname{cosec}\theta = \cot\theta + \sqrt{3}$

$$\Rightarrow \frac{1}{\sin\theta} = \frac{\cos\theta}{\sin\theta} + \sqrt{3}$$

$$\Rightarrow 1 = \cos\theta + \sqrt{3}\sin\theta \qquad \{\because \sin\theta \neq 0\}$$

$$\Rightarrow \frac{1}{2}\cos\theta + \frac{\sqrt{3}}{2}\sin\theta = \frac{1}{2}$$

$$\Rightarrow \cos\left(\theta - \frac{\pi}{3}\right) = \cos\frac{\pi}{3}$$

$$\Rightarrow \theta - \frac{\pi}{3} = 2n\pi \pm \frac{\pi}{3}$$

$$\Rightarrow \theta = 2n\pi \quad \text{or} \quad \theta = 2n\pi + \frac{2\pi}{3}$$

$\because \sin\theta \neq 0$

$$\Rightarrow \theta = 2n\pi + \frac{2\pi}{3}; n \in I$$

29. $\operatorname{cosec} x = 1 + \cot x$

Ans. $2n\pi + \frac{\pi}{2}; n \in I$

Sol. Given, $\operatorname{cosec} x = 1 + \cot x$

$$\Rightarrow \frac{1}{\sin x} = \frac{\sin x + \cos x}{\sin x}$$

$$\Rightarrow 1 = \sin x + \cos x \qquad \{\because \sin x \neq 0\}$$

$$\Rightarrow \frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{2}}\cos x = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \cos\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}},$$

$\Rightarrow x - \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4}; n \in I$

$\Rightarrow x = 2n\pi$ or $x = 2n\pi + \frac{\pi}{2}; n \in I$

$\because \sin x \neq 0$

$\Rightarrow x = 2n\pi + \frac{\pi}{2}; n \in I$

30. $(2 + \sqrt{3}) \cos\theta = 1 - \sin\theta$

Ans. $2n\pi + \frac{\pi}{2}$ or $2n\pi - \frac{\pi}{3}; n \in I$

Sol. Given, $\sin\theta = 1 - (2 + \sqrt{3}) \cos\theta$

$\Rightarrow \sin^2\theta = 1 + \left(4 + 3 + 4\sqrt{3}\right)\cos^2\theta - 2\left(2 + \sqrt{3}\right)\cos\theta$

$\Rightarrow 1 - \cos^2\theta = 1 + \left(7 + 4\sqrt{3}\right)\cos^2\theta - 2\left(2 + \sqrt{3}\right)\cos\theta$

$\Rightarrow \left(8 + 4\sqrt{3}\right)\cos^2\theta - 2\left(2 + \sqrt{3}\right)\cos\theta = 0$

$\Rightarrow 4\left(2 + \sqrt{3}\right)\cos^2\theta - 2\left(2 + \sqrt{3}\right)\cos\theta = 0$

$\Rightarrow 2\cos^2\theta - \cos\theta = 0$

$\Rightarrow \cos\theta\,[2\cos\theta - 1] = 0$

Now either $\cos\theta = 0$, or $2\cos\theta - 1 = 0$

If $\cos\theta = 0$ then $\cos\theta = \cos\frac{\pi}{2}$

$\Rightarrow \theta = 2n\pi + \frac{\pi}{2}; n \in I$

If $\cos\theta = \frac{1}{2} = \cos\frac{\pi}{3}$

Then $\theta = 2n\pi - \frac{\pi}{3}; n \in I$

31. $\tan\theta + \sec\theta = \sqrt{3}$

Ans. $2n\pi+\frac{\pi}{6};\ n\in I$

Sol. Given, $\tan\theta+\sec\theta=\sqrt{3}$

$\Rightarrow \frac{\sin\theta}{\cos\theta}+\frac{1}{\cos\theta}=\sqrt{3}$

$\Rightarrow \sqrt{3}\cos\theta-\sin\theta=1 \qquad \{\because \cos\theta\neq 0\}$

$\Rightarrow \frac{\sqrt{3}}{2}\cos\theta-\frac{1}{2}\sin\theta=\frac{1}{2}$

$\Rightarrow \cos 30^\circ\cos\theta-\sin 30^\circ\sin\theta=\frac{1}{2}$

$\Rightarrow \cos\left(\theta+\frac{\pi}{6}\right)=\cos\frac{\pi}{3}$

$\Rightarrow \theta+\frac{\pi}{6}=2n\pi\pm\frac{\pi}{3};\ n\in I$

$\Rightarrow \theta=2n\pi-\frac{\pi}{3}-\frac{\pi}{6};\ n\in I \qquad$ or $\qquad \theta=2n\pi+\frac{\pi}{3}-\frac{\pi}{6}$

$\because \cos\theta\neq 0$

$\Rightarrow \theta=2n\pi+\frac{\pi}{6};n\in I$

32. $\sec\theta-1=(\sqrt{2}-1)\tan\theta$

Ans. $\theta=2n\pi$ or $2n\pi+\frac{\pi}{4};\ n\in I$

Sol. $\sec\theta-1=\left(\sqrt{2}-1\right)\tan\theta$

$\Rightarrow 1-\cos\theta=\left(\sqrt{2}-1\right)\sin\theta \qquad (\because\ \cos\theta\neq 0)$

$\Rightarrow \left(\sqrt{2}-1\right)\sin\theta+\cos\theta=1$

$\sqrt{2}\sin\theta+\cos\theta=\sin\theta$

$\Rightarrow (1-\cos\theta)\cos\frac{\pi}{8}=\sin\frac{\pi}{8}\sin\theta$

$\Rightarrow \sin\frac{\pi}{8}\sin\theta+\cos\theta\cos\frac{\pi}{8}=\cos\frac{\pi}{8}$

$$\Rightarrow \cos\left(\theta - \frac{\pi}{8}\right) = \cos\left(\frac{\pi}{8}\right)$$

$$\Rightarrow \theta - \frac{\pi}{8} = 2m\pi \pm \frac{\pi}{8}$$

$$\Rightarrow \theta = 2n\pi; n \in I \qquad \text{or} \qquad \theta = 2n\pi + \frac{\pi}{4}; n \in I$$

1. $\sin\theta + \sin 7\theta = \sin 4\theta$

Ans. $\frac{n\pi}{4}$ or $\frac{1}{3}\left(2n\pi \pm \frac{\pi}{3}\right); n \in I$

Sol. $2\sin 4\theta \cos 3\theta - \sin 4\theta = 0$

$\Rightarrow \sin 4\theta (2\cos 3\theta - 1) = 0$

$\sin 4\theta = 0$ or $2\cos 3\theta - 1 = 0$

$\Rightarrow 4\theta = n\pi$ or $3\theta = 2n\pi \pm \frac{\pi}{3}; n \in I$

2. $\cos\theta + \cos 7\theta = \cos 4\theta$

Ans. $\left(n + \frac{1}{2}\right)\frac{\pi}{4}$ or $\left(2n \pm \frac{1}{3}\right)\frac{\pi}{3}; n \in I$

Sol. $2\cos 4\theta \cos 3\theta - \cos 4\theta = 0$

$\Rightarrow \cos 4\theta (2\cos 3\theta - 1) = 0$

$\Rightarrow \cos 4\theta = 0$ or $\cos 3\theta = \frac{1}{2}$

$\Rightarrow 4\theta = n\pi + \frac{\pi}{2}$ or $3\theta = 2n\pi \pm \frac{\pi}{3}; n \in I$

3. $\cos\theta + \cos 3\theta = 2\cos 2\theta$

Ans. $\left(n + \frac{1}{2}\right)\frac{\pi}{2}$ or $2n\pi; n \in I$

Sol. $2\cos 2\theta \cos\theta - 2\cos 2\theta = 0$

$\Rightarrow 2\cos 2\theta (\cos\theta - 1) = 0$

$\Rightarrow 2\cos 2\theta = 0$ or $\cos\theta - 1 = 0$

$\Rightarrow 2\theta = n\pi + \frac{\pi}{2}$ or $\theta = 2n\pi; n \in I$

4. $\sin 4\theta - \sin 2\theta = \cos 3\theta$

Ans. $\left(n + \frac{1}{2}\right)\frac{\pi}{3}$ or $n\pi + (-1)^n \frac{\pi}{6}$

Sol. $2\cos 3\theta \sin\theta - \cos 3\theta = 0$

$\Rightarrow \cos 3\theta\,(2\sin\theta - 1) = 0$

$\Rightarrow \cos 3\theta = 0$ or $2\sin\theta - 1 = 0$

$\Rightarrow 3\theta = n\pi + \frac{\pi}{2}; n \in I$ or $\theta = n\pi + (-1)^n \frac{\pi}{6}; n \in I$

5. $\cos\theta - \sin 3\theta = \cos 2\theta$

Ans. $\frac{2n\pi}{3}$ or $\left(n + \frac{1}{4}\right)\pi$ or $\left(2n - \frac{1}{2}\right)\pi$

Sol. $\cos\theta - \cos 2\theta - \sin 3\theta = 0$

$\Rightarrow 2\sin\frac{3\theta}{2}\sin\frac{\theta}{2} - 2\sin\frac{3\theta}{2}\cos\frac{3\theta}{2} = 0$

$\Rightarrow 2\sin\frac{3\theta}{2}\left\{\sin\frac{\theta}{2} - \cos\frac{3\theta}{2}\right\} = 0$

$\Rightarrow 2\sin = \frac{3\theta}{2} = 0,$ or $\sin\frac{\theta}{2} - \cos\frac{3\theta}{2} = 0$

$\Rightarrow \frac{3\theta}{2} = n\pi$ or $\frac{3\theta}{2} = 2n\pi \pm \left(\frac{\pi}{2} - \frac{\theta}{2}\right)$

$\Rightarrow \theta = \frac{2n\pi}{3}; n \in I$ or $\theta = \left(n + \frac{1}{4}\right); n \in I$ or $\theta = \left(2n - \frac{1}{2}\right)\pi; n \in I$

6. $\sin 7\theta = \sin\theta + \sin 3\theta$

Ans. $\frac{n\pi}{3}$ or $\left(2n \pm \frac{1}{3}\right)\frac{\pi}{4}$

Sol. $\sin\theta + \sin 3\theta - \sin 7\theta = 0$

$\Rightarrow -2\cos 4\theta \sin 3\theta + \sin 3\theta = 0$

$\Rightarrow -\sin 3\theta\,(2\cos 4\theta - 1) = 0$

$\Rightarrow 2\sin 3\theta = 0,$ or $2\cos 4\theta - 1 = 0$

$\Rightarrow 3\theta = n\pi$ or $4\theta = 2n\pi \pm \frac{\pi}{3}$

7. $\cos\theta + \cos 2\theta + \cos 3\theta = 0$

Ans. $\left(n + \frac{1}{2}\right)\frac{\pi}{2}; n \in I$ or $2n\pi \pm \frac{2\pi}{3}$

Sol. $2\cos 2\theta \cos\theta + \cos 2\theta = 0$

$\Rightarrow \cos 2\theta\,\{2\cos\theta + 1) = 0$

$\Rightarrow \cos 2\theta = 0,$ or $2\cos\theta + 1 = 0$

$\Rightarrow 2\theta = (2n+1)\frac{\pi}{2}$ or $\theta = 2n\pi \pm \frac{2\pi}{3}; n \in I$

8. $\sin\theta + \sin 3\theta + \sin 5\theta = 0$

Ans. $\frac{n\pi}{3}$ or $\left(n \pm \frac{1}{3}\right)\pi$

Sol. $\sin\theta + \sin 5\theta + \sin 3\theta = 0$

$\Rightarrow 2\sin 3\theta \cos 2\theta + \sin 3\theta = 0$

$\Rightarrow \sin 3\theta\,(2\cos 2\theta + 1) = 0$

$\Rightarrow \sin 3\theta = 0$ or $\cos 2\theta = \frac{-1}{2}$

$\Rightarrow 3\theta = n\pi; n \in I$ or $\cos 2\theta = \cos\left(\frac{2\pi}{3}\right)$

$\Rightarrow \theta = \frac{n\pi}{3}; n \in I$ or $2\theta = 2n\pi \pm \frac{\pi}{3}; n \in I$

$\Rightarrow \theta = \frac{n\pi}{3}; n \in I$ or

9. $\sin 2\theta - \cos 2\theta - \sin\theta + \cos\theta = 0$

Ans. $2n\pi$ or $\left(\frac{2n}{3} + \frac{1}{2}\right)\pi$

Sol. $\sin 2\theta - \cos 2\theta - \sin\theta + \cos\theta = 0$

$\Rightarrow \sin 2\theta - \sin\theta + \cos\theta - \cos 2\theta = 0$

$\Rightarrow 2\cos\left(\frac{3\theta}{2}\right)\sin\left(\frac{\theta}{2}\right) + 2\sin\left(\frac{3\theta}{2}\right)\sin\left(\frac{\theta}{2}\right) = 0$

$\Rightarrow 2\sin\frac{\theta}{2}\left(\cos\frac{3\theta}{2} + \sin\frac{3\theta}{2}\right) = 0$

$\Rightarrow \sin\frac{\theta}{2} = 0$ or $\sin\frac{3\theta}{2} = -\cos\left(\frac{3\theta}{2}\right)$

$\Rightarrow \frac{\theta}{2} = n\pi; n \in I$ or $\frac{3\theta}{2} = 2n\pi \pm \left(\frac{\pi}{2} + \frac{3\theta}{2}\right)$

$\Rightarrow \theta = 2n\pi; n \in I$ or $\theta = \left(\frac{2n\pi}{3} - \frac{\pi}{6}\right); n \in I$

10. $\sin(3\theta + \alpha) + \sin(3\theta - \alpha) + \sin(\alpha - \theta) - \sin(\alpha + \theta) = \cos\alpha$

Ans. $n\pi + (-1)^n \frac{\pi}{6}$ or $n\pi + (-1)^n \frac{\pi}{10}$ or $n\pi - (-1)^n \frac{3\pi}{10}$

Sol. $2 \sin 3\theta \cos \alpha + 2 \cos \alpha \sin(-\theta) - \cos \alpha = 0$

$\Rightarrow \cos \alpha \{2 \sin 3\theta - 2 \sin \theta - 1\} = 0$

$\cos \alpha = 0$ or $2 \sin 3\theta - 2 \sin \theta = 1$

$\Rightarrow \sin \theta = \frac{1}{2}$ or $4\sin^2 \theta + 2\sin \theta - 1 = 0$

$\Rightarrow \sin \theta = \frac{1}{2}$ or $\sin \theta = \frac{\sqrt{5}-1}{4}$ or $\sin \theta = -\frac{\left(\sqrt{5}+1\right)}{4}$

$\Rightarrow \theta = n\pi + (-1)^n \frac{\pi}{6}; n \in I$ or $\theta = n\pi + (-1)^n \frac{\pi}{10}; n \in I$ or $\theta = n\pi + (-1)^{n+1} \frac{3\pi}{10}; n \in I$

11. $\cos(3\theta + \alpha)\cos(3\theta - \alpha) + \cos(5\theta + \alpha)\cos(5\theta - \alpha) = \cos 2\alpha$

Ans. $\left(n + \frac{1}{2}\right)\frac{\pi}{8}$ or $\left(n + \frac{1}{2}\right)\frac{\pi}{2}$

Sol. $\frac{1}{2}\{\cos 6\theta + \cos 2\alpha + \cos 10\theta + \cos 2\alpha\} = \cos 2\alpha$

$\Rightarrow \cos 6\theta + \cos 10\theta + 2\cos 2\alpha = 2\cos 2\alpha$

$\Rightarrow \cos 6\theta + \cos 10\theta = 0$

$\Rightarrow \cos 8\theta = 0$ or $\cos 2\theta = 0$

$\Rightarrow 8\theta = n\pi + \frac{\pi}{2}; n \in I$ or $2\theta = n\pi + \frac{\pi}{2}; n \in I$

$\Rightarrow \theta = \left(n + \frac{1}{2}\right)\frac{\pi}{8}; n \in I$ or $\theta = \left(n + \frac{1}{2}\right)\frac{\pi}{2}; n \in I$

12. $\cos n\theta = \cos(n-2)\theta + \sin \theta$

Ans. $m\pi$ or $\frac{1}{n-1}\left[m\pi - (-1)^m \frac{\pi}{6}\right]$

Sol. $\cos n\theta = \cos(n-2)\theta + \sin \theta$

$\Rightarrow 2\sin(n\theta - \theta)\sin(-\theta) - \sin \theta = 0$

$\Rightarrow \sin \theta \{2 \sin(n\theta - \theta) + 1\} = 0$

$\Rightarrow \sin \theta = 0,$ or $2 \sin(n\theta - \theta) + 1 = 0$

$\Rightarrow \theta = m\pi; m \in I$ or $\theta = \frac{1}{n-1}\left(m\pi + (-1)^{m+1} \frac{\pi}{6}\right); m \in I$

13. $\sin\frac{n+1}{2}\theta = \sin\frac{n-1}{2}\theta + \sin\theta$

Ans. $2m\pi$ or $\frac{4m\pi}{n+1}$ or $\frac{4m\pi}{n-1}$

Sol. $\sin\left(\frac{(n+1)}{2}\theta\right) - \sin\left(\frac{(n-1)}{2}\theta\right) = \sin\theta$

$\Rightarrow 2\cos\left(\frac{n\theta}{2}\right)\sin\left(\frac{\theta}{2}\right) = \sin\theta$

$\Rightarrow 2\sin\left(\frac{\theta}{2}\right)\left[\cos\left(\frac{n\theta}{2}\right) - \cos\frac{\theta}{2}\right] = 0$

$\Rightarrow \sin\frac{\theta}{2} = 0$ or $\cos\left(\frac{n\theta}{2}\right) = \cos\left(\frac{\theta}{2}\right)$

$\Rightarrow \frac{\theta}{2} = m\pi; m \in I$ or $n\theta = 4m\pi \pm \theta; m \in I$

14. $\sin m\theta + \sin n\theta = 0$

Ans. $\frac{2r\pi}{m+n}$ or $(2r+1)\frac{\pi}{m-n}$

Sol. $\sin(m\theta) + \sin n\theta = 0$

$\Rightarrow 2\sin\left(\frac{(m+n)\theta}{2}\right)\cos\left(\frac{(m-n)\theta}{2}\right) = 0$

$\Rightarrow \sin\left(\frac{(m+n)\theta}{2}\right) = 0$ or $\cos\left(\frac{(m-n)\theta}{2}\right) = 0$

$\Rightarrow \theta = \frac{2r\pi}{m+n}; r \in I$ or $\theta = \frac{(2r+1)\pi}{(m-n)}; r \in I$

15. $\cos m\theta + \cos n\theta = 0$

Ans. $(2r+1)\frac{\pi}{m \pm n}$

Sol. $2\cos\frac{m\theta + n\theta}{2}\cos\frac{m\theta - n\theta}{2} = 0$

$\Rightarrow \cos\frac{\theta(m+n)}{2}\cos\frac{\theta(m-n)}{2}=0$

$\Rightarrow \cos\frac{(m+n)\theta}{2}=0$ or $\cos\frac{(m-n)\theta}{2}=0$

$\Rightarrow \theta=\frac{(2r+1)\pi}{m+n};r\in I$ or $\theta=\frac{(2r+1)\pi}{m-n};r\in I$

16. $\sin^2 n\theta-\sin^2(n-1)\theta=\sin^2\theta$

Ans. $m\pi$ or $\frac{m\pi}{n-1}$ or $\left(m+\frac{1}{2}\right)\frac{\pi}{n}$

Sol. $\sin^2 n\theta-\sin^2\theta-\sin^2(n-1)\theta=0$

$\Rightarrow \sin(n+1)\theta\sin(n-1)\theta-\sin(n-1)\theta=0$

$\Rightarrow \sin(n-1)\theta\{\sin(n+1)\theta-\sin(n-1)\theta\}=0$

$\Rightarrow \sin(n-1)\theta.2\cos n\theta\sin\theta=0$

$\Rightarrow \sin(n-1)\theta=0$ or $\cos n\theta=0$ or $\sin\theta=0$

$\Rightarrow \theta=m\pi;m\in I$ or $\theta=(2m+1)\frac{\pi}{2n};m\in I$ or $\theta=\frac{m\pi}{n-1};m\in I$

17. $\sin 3\theta+\cos 2\theta=0$

Ans. $2n\pi-\frac{\pi}{2};\frac{1}{5}\left(2n\pi-\frac{\pi}{2}\right)$

Sol. $\cos 2\theta-\cos\left(\frac{\pi}{2}+3\theta\right)=0$

$\Rightarrow 2\sin\left(\frac{\pi}{4}+\frac{5\theta}{2}\right)\sin\left(\frac{\pi}{4}+\frac{\theta}{2}\right)=0$

$\Rightarrow \sin\left(\frac{\pi}{4}+\frac{5\theta}{2}\right)=0$ or $\sin\left(\frac{\pi}{4}+\frac{\theta}{2}\right)=0$

$\Rightarrow \frac{\pi}{4}+\frac{5\theta}{2}=n\pi;n\in I$ or $\frac{\pi}{4}+\frac{\theta}{2}=n\pi;n\in I$

$\Rightarrow \theta=\left(2n\pi-\frac{\pi}{2}\right)\frac{1}{5};n\in I$ or $\theta=\left(2n\pi-\frac{\pi}{2}\right);n\in I$

18. $4\cot 2\theta=\cot^2\theta-\tan^2\theta$

Ans. $n\pi\pm\frac{\pi}{4}$

Sol. Given $4\cot 2\theta=\cot^2\theta-\tan^2\theta$

$$\Rightarrow \frac{4}{\tan 2\theta} = \left(\frac{1}{\tan^2\theta} - \tan^2\theta \right)$$

$$\Rightarrow \frac{4\left(1-\tan^2\theta\right)}{2\tan\ \theta} - \left(\frac{1-\tan^4\theta}{\tan^2\theta} \right) = 0$$

$$\Rightarrow \left(1-\tan^2\theta\right)\left[\frac{2}{\tan\theta} - \frac{1+\tan^2\theta}{\tan^2\theta} \right]$$

$\Rightarrow (1 - \tan^2\theta)(1 - \tan\theta)^2 = 0$

$\Rightarrow 1 - \tan^2\theta = 0$ or $(1 - \tan\theta)^2 = 0$

$\Rightarrow \tan\theta = 1$ or $\Rightarrow \tan\theta = -1$

$$\Rightarrow \theta = n\pi \pm \frac{\pi}{4}; n \in I$$

19. $3\tan(\theta - 15^\circ) = \tan(\theta + 15^\circ)$

Ans. $n\pi + \frac{\pi}{4}$

Sol. Given $3\tan(\theta - 15^\circ) = \tan(\theta + 15^\circ)$

$$\Rightarrow \frac{3\sin\left(\theta-15^\circ\right)}{\cos\left(\theta-15^\circ\right)} = \frac{\sin\left(\theta+15^\circ\right)}{\cos\left(\theta+15^\circ\right)}$$

$\Rightarrow 3\sin(\theta - 15^\circ)\cos(\theta + 15^\circ) = \sin(\theta + 15^\circ)\cos(\theta - 15^\circ)$

$\Rightarrow 3\ \{\sin 2\theta - \sin 30^\circ\} = \sin 2\theta + \sin 30^\circ$

$$\Rightarrow 2\sin 2\theta = 4\sin 30^\circ = 4 \times \frac{1}{2} = 2$$

$\Rightarrow \sin 2\theta = 1$

$$\Rightarrow (4n+1)\frac{\pi}{2} = 2\theta$$

$$\Rightarrow 2n\pi + \frac{\pi}{2} = 2\theta$$

$$\Rightarrow \theta = n\pi + \frac{\pi}{4}; n \in I$$

20. $\tan\theta + \tan 2\theta + \tan 3\theta = 0$

Ans. $\theta = n\pi$ or $n\pi \pm \frac{\pi}{3}$; also $\theta = n\pi \pm \alpha/2$, where $\cos\alpha = 1/3$

Sol. $\tan\theta+\tan 2\theta+\tan 3\theta=0$

$$\tan\theta+\frac{2\tan\theta}{1-\tan^2\theta}+\frac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}=0$$

$$\Rightarrow \tan\theta=0 \quad \text{or} \quad 1+\frac{2}{1-\tan^2\theta}+\frac{3-\tan^2\theta}{1-3\tan^2\theta}=0$$

$$\Rightarrow \tan\theta=0 \quad \text{or} \quad \tan^2\theta=3 \quad \text{or} \quad \tan^2\theta=\frac{1}{2}$$

$$\Rightarrow \tan\theta=0 \quad \text{or} \quad \tan^2\theta=\left(\sqrt{3}\right)^2 \quad \text{or} \quad \tan^2\theta=\left(\frac{1}{\sqrt{2}}\right)^2$$

$$\Rightarrow \theta=n\pi \quad \text{or} \quad \theta=n\pi\pm\frac{\pi}{3} \quad \text{or} \quad \theta=n\pi\pm\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

21. $\tan\theta+\tan 2\theta+\sqrt{3}\tan\theta\tan 2\theta=\sqrt{3}$

Ans. $\left(n+\frac{1}{3}\right)\frac{\pi}{3}$

Sol. $\tan\theta+\tan 2\theta=\sqrt{3}\left(1-\tan\theta\tan 2\theta\right)$

$$\Rightarrow \frac{\tan\theta+\tan 2\theta}{1-\tan\theta\tan 2\theta}=\sqrt{3}$$

$$\Rightarrow \tan\left(\theta+2\theta\right)=\sqrt{3}$$

$$\Rightarrow \tan 3\theta=\sqrt{3}$$

$$\Rightarrow 3\theta=n\pi+\frac{\pi}{3};n\in I$$

$$\Rightarrow \theta=\frac{1}{3}\left(n\pi+\frac{\pi}{3}\right);n\in I$$

22. $\sin 3\alpha=4\sin\alpha\sin(x+\alpha)\sin(x-\alpha)\ (\alpha\neq n\pi)$

Ans. $n\pi\pm\frac{\pi}{3}$

Sol. $\sin 3\alpha=4\sin\alpha\sin\left(x+\alpha\right)\sin\left(x-\alpha\right);\left(\alpha\neq n\pi\right)$

$$\Rightarrow \sin 3\alpha=4\sin\alpha\left(\sin^2 x-\sin^2\alpha\right)$$

$$\Rightarrow 3\sin\alpha-4\sin^3\alpha=4\sin\alpha\sin^2 x-4\sin^3\alpha$$

$\Rightarrow \sin\alpha\left(4\sin^2 x - 3\right) = 0$

$\Rightarrow \sin\alpha = 0$ or $\sin^2 x = \left(\frac{\sqrt{3}}{2}\right)^2 = \sin^2\frac{\pi}{3}$

$\because \alpha \neq n\pi$

$\Rightarrow x = n\pi \pm \frac{\pi}{3}; n \in I$

23. Prove that the equation $x^3 - 2x + 1 = 0$ is satisfied by putting for x, either of the values

$\sqrt{2}\ \sin 45^\circ, 2\sin 18^\circ$ and $2\sin 234^\circ$

सिद्ध कीजिये कि समीकरण $x^3 - 2x + 1 = 0$, x का मान $\sqrt{2}\ \sin 45^\circ, 2\sin 18^\circ$ तथा $2\sin 234^\circ$ रखने से सन्तुष्ट होता है।

Ans.

Sol. $x^3 - 2x + 1 = 0$

$\Rightarrow (x - 1)(x^2 + x - 1) = 0$

$\Rightarrow x = 1$ or $x = \frac{\sqrt{5}-1}{2}$ or $x = -\left(\frac{\sqrt{5}+1}{2}\right)$

$\Rightarrow x = \sqrt{2}\sin 45^\circ$ or $x = 2\sin 18^\circ$ or $x = 2\sin 234^\circ$

24. $\sin 9\theta = \sin\theta$

Ans. $\frac{n\pi}{4}$ or $\frac{(2n+1)\pi}{10}$

Sol. $\sin 9\theta = \sin\theta$

$\Rightarrow 2\cos 5\theta \sin 4\theta = 0$

$\Rightarrow \cos 5\theta = 0$ or $\sin 4\theta = 0$

$\Rightarrow \theta = (2n+1)\frac{\pi}{10}; n \in I$ or $\theta = \frac{n\pi}{4}; n \in I$

25. $\sin 3\theta = \sin 2\theta$

Ans. $2n\pi$ or $\frac{(2n+1)\pi}{5}$

Sol. $\sin 3\theta = \sin 2\theta$

$$\Rightarrow 2\cos\left(\frac{5\theta}{2}\right)\sin\left(\frac{\theta}{2}\right)=0$$

$$\Rightarrow \sin\frac{\theta}{2}=0 \qquad \text{or} \qquad \cos\frac{5\theta}{2}=0$$

$$\Rightarrow \theta=2n\pi;\, n\in I \qquad \text{or} \qquad \theta=(2n+1)\frac{\pi}{5};\, n\in I$$

26. $\cos m\theta=\cos n\theta$

Ans. $\frac{2r\pi}{m-n}$ or $\frac{2r\pi}{m+n}$

Sol. $\cos m\theta=\cos n\theta$

$$\Rightarrow 2\sin\left(\frac{(m+n)\theta}{2}\right)\sin\left(\frac{(n-m)\theta}{2}\right)=0$$

$$\Rightarrow \sin\left(\frac{(m+n)\theta}{2}\right)=0 \qquad \text{or} \qquad \sin\left(\frac{(n-m)\theta}{2}\right)=0$$

$$\Rightarrow \theta=\frac{2r\pi}{m+n};\, r\in I \qquad \text{or} \qquad \theta=\frac{2r\pi}{n-m};\, r\in I$$

27. $\sin 2\theta=\cos 3\theta$

Ans. $\left(2n+\frac{1}{2}\right)\frac{\pi}{5}$ or $2n\pi-\frac{\pi}{2}$

Sol. $\sin 2\theta=\cos 3\theta$

$$\Rightarrow \cos\left(\frac{\pi}{2}-2\theta\right)=\cos 3\theta$$

$$\Rightarrow 3\theta=2n\pi\pm\frac{\pi}{2}-2\theta$$

$$\Rightarrow \theta=\left(2n+\frac{1}{2}\right)\frac{\pi}{5};\, n\in I \qquad \text{or} \qquad \theta=2n\pi-\frac{\pi}{2};\, n\in I$$

28. $\cos 5\theta=\cos 4\theta$

Ans. $2n\pi$ or $\frac{2n\pi}{9}$

Sol. $\cos 5\theta = \cos 4\theta$

$\Rightarrow 5\theta = 2n\pi \pm 4\theta$

$\Rightarrow \theta = 2n\pi; n \in I$ or $\theta = \frac{2n\pi}{9}; n \in I$

29. $\cos m\theta = \sin n\theta$

Ans. $\left(2r + \frac{1}{2}\right)\frac{\pi}{m+n}$ or $\left(2r - \frac{1}{2}\right)\frac{\pi}{m-n}$

Sol. $\cos m\theta = \sin n\theta$

$\Rightarrow \cos m\theta = \cos\left(\frac{\pi}{2} - n\theta\right)$

$\Rightarrow m\theta = 2r\pi \pm \frac{\pi}{2} - n\theta$

$\Rightarrow \theta = \left(2r + \frac{1}{2}\right)\frac{\pi}{m+n}; r \in I$ or $\theta = \left(2r - \frac{1}{2}\right)\frac{\pi}{m-n}; r \in I$

30. $\cot\theta = \tan 8\theta$

Ans. $\left(n + \frac{1}{2}\right)\frac{\pi}{9}$

Sol. $\cot\theta = \tan 8\theta$

$\Rightarrow \tan\left(\frac{\pi}{2} - \theta\right) = \tan 8\theta$

$\Rightarrow 8\theta = n\pi + \frac{\pi}{2} - \theta$

$\Rightarrow \theta = \left(n + \frac{1}{2}\right)\frac{\pi}{9}; n \in I$

31. $\cot\theta = \tan n\theta$

Ans. $\left(m+\frac{1}{2}\right)\frac{\pi}{n+1}$

Sol. $\cot\theta = \tan n\theta$

$\Rightarrow n\theta = n\pi + \frac{\pi}{2} - \theta$

$\Rightarrow \theta = \left(m+\frac{1}{2}\right)\frac{\pi}{n+1}; m \in I$

32. $\tan 2\theta = \tan\frac{2}{\theta}$

Ans. $\frac{n\pi}{4} \pm \sqrt{1+\frac{n^2\pi^2}{16}}$

Sol. $\tan 2\theta = \tan\left(\frac{2}{\theta}\right)$

$\Rightarrow 2\theta = n\pi + \frac{2}{\theta}$

$\Rightarrow 2\theta^2 - n\pi\theta - 2 = 0$

$\Rightarrow \theta = \frac{n\pi}{4} \pm \sqrt{1+\frac{n^2\pi^2}{16}}$

33. $\tan 2\theta \tan\theta = 1$

Ans. $n\pi \pm \frac{\pi}{6}$

Sol. $\tan 2\theta \tan\theta = 1$

$\Rightarrow \frac{2\tan^2\theta}{1-\tan^2\theta} = 1 \Rightarrow 3\tan^2\theta - 1 = 0$

$\Rightarrow \tan^2\theta = \tan^2\frac{\pi}{6}$

$\Rightarrow \theta = n\pi \pm \frac{\pi}{6}; n \in I$

34. $\tan^2 3\theta = \cot^2 \alpha$

Ans. $\left(n+\frac{1}{2}\right)\frac{\pi}{3} \pm \frac{\alpha}{3}$

Sol. $\tan^2 3\theta = \cot^2 \alpha$

$\Rightarrow \cot^2\left(\frac{\pi}{2} - 3\theta\right) = \cot^2 \alpha$

$\Rightarrow 3\theta = n\pi + \frac{\pi}{2} \pm \alpha$

$\Rightarrow \theta = \left(n+\frac{1}{2}\right)\frac{\pi}{3} \pm \frac{\alpha}{3}$

35. $\tan 3\theta = \cot \theta$

Ans. $\left(n+\frac{1}{2}\right)\frac{\pi}{4}$

Sol. $\tan 3\theta = \cot \theta$

$\Rightarrow \tan 3\theta = \tan\left(\frac{\pi}{2} - \theta\right)$

$\Rightarrow \theta = \frac{1}{4}\left(n\pi + \frac{\pi}{2}\right)$

36. $\tan^2 3\theta = \tan^2 \alpha$

Ans. $\frac{n\pi}{3} \pm \frac{\alpha}{3}$

Sol. $\tan^2 3\theta = \tan^2 \alpha$

$\Rightarrow 3\theta = n\pi \pm \alpha$

$\Rightarrow \theta = \frac{n\pi}{3} \pm \frac{\alpha}{3}$

37. $\tan mx + \cot nx = 0$

Ans. $\left(r+\frac{1}{2}\right)\frac{\pi}{m-n}$

Sol. $\tan mx + \cot nx = 0$

$\Rightarrow \tan mx = -\cot(nx)$

$\Rightarrow \tan mx = \tan\left(\frac{\pi}{2} + nx\right)$

$\Rightarrow mx = r\pi + \frac{\pi}{2} + nx$

$\Rightarrow x = \left(r + \frac{1}{2}\right)\frac{\pi}{m-n}; r \in I$

38. $\tan(\pi \cot\theta) = \cot(\pi \tan\theta)$

Ans. $\tan\theta = \frac{2n+1 \pm \sqrt{4n^2+4n-15}}{4}$, where $n > 1$ or < -2

Sol. $\tan(\pi\cot\theta) = \cot(\pi\tan\theta)$

$\Rightarrow \tan(\pi\cot\theta) = \tan\left(\frac{\pi}{2} - \pi\tan\theta\right)$

$\Rightarrow \pi\cot\theta = n\pi + \frac{\pi}{2} - \pi\tan\theta$

$\Rightarrow \cot\theta + \tan\theta = n + \frac{1}{2}$

$\Rightarrow \frac{\tan^2\theta + 1}{\tan\theta} = \frac{2n+1}{2}$

$\Rightarrow 2\tan^2\theta + 2 - (2n+1)\tan\theta = 0$

$\tan\theta = \frac{(2n+1) \pm \sqrt{4n^2+4n-15}}{4}$

$\because \tan\theta + \cot\theta \geq 2$ or $\tan\theta + \cot\theta \leq -2$

$\Rightarrow n \geq \frac{3}{2}$ or $n \leq -\frac{5}{2}$

$\Rightarrow n > 1$ or $n < -2$

39. Find all the angles between 0° and 90° which satisfy the equation $\sec^2\theta \operatorname{cosec}^2\theta + 2\operatorname{cosec}^2\theta = 8$

0° तथा 90° के बीच वो कोण ज्ञात करो जबकि समीकरण $\sec^2\theta \operatorname{cosec}^2\theta + 2\operatorname{cosec}^2\theta = 8$ है।

Ans. 45° and 60°

Sol. $\sec^2\theta \operatorname{cosec}^2\theta + 2\operatorname{cosec}^2\theta = 8$

$\Rightarrow 8\cos^2\theta \sin^2\theta = 1 + 2\cos^2\theta$

$\Rightarrow 8\cos^4\theta - 6\cos^2\theta + 1 = 0$

$\Rightarrow \cos^2\theta = \frac{1}{2} = \cos^2\frac{\pi}{4}$ or $\cos^2\theta = \frac{1}{4} = \cos^2\frac{\pi}{3}$

$\Rightarrow \theta = 45^\circ$ or $\theta = 60^\circ$